Direction of Arrival (DOA) Estimation Methods
IN5450 Project 2

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Agenda

1. Signal Model & Setup
2. Correlation Matrix
3. Conventional Beamformer - DAS
4. Capon / MVDR Beamformer
5. Eigenvalue Decomposition
6. MUSIC Algorithm
7. Eigenvector Method
8. Incorrect Source Count

Signal model

Setup (from `generate_data.m`):

- $M = 10$ sensors, ULA with $d = 0.5\lambda$
- $N = 100$ snapshots
- Two incoherent sources: $\theta_1 = 0^\circ$, $\theta_2 = -10^\circ$
- Both at $\mathrm{SNR} = 0\,\mathrm{dB}$, relative phase $\gamma = 45^\circ$
- Additive complex Gaussian noise ($\sigma^2 = 2$)

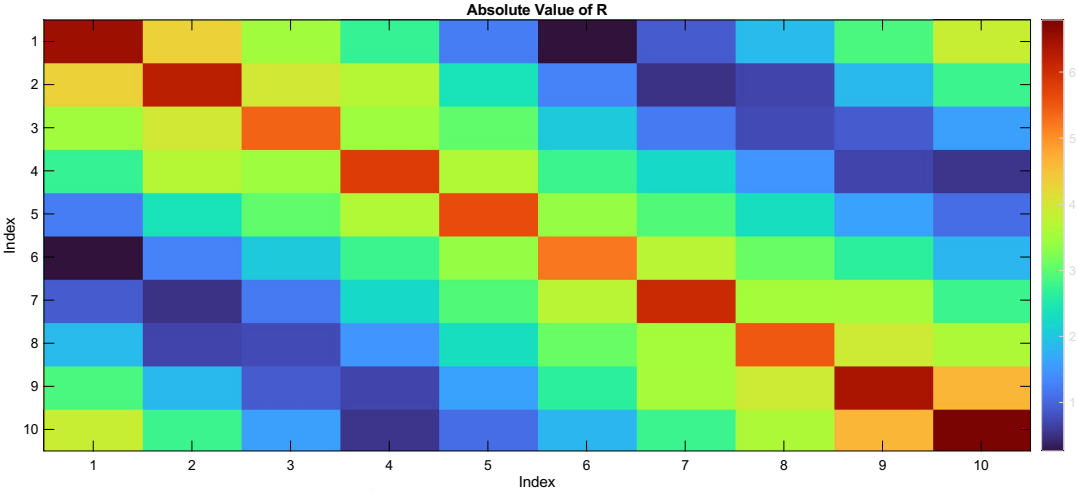
Data model:

$$\mathrm{x}(t) = \mathrm{A}\,\mathrm{s}(t) + \mathrm{n}(t), \qquad \mathrm{A} = [\mathrm{a}(\theta_1)\ \ \mathrm{a}(\theta_2)], \qquad [\mathrm{a}(\theta)]_m = e^{-jmkd\sin\theta}$$

Sample correlation matrix:

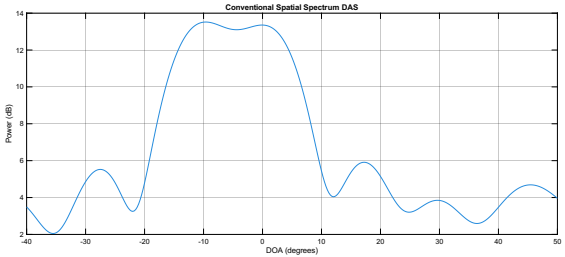
$$\hat{\mathrm{R}} = \frac{1}{N}\sum_{t=1}^{N} \mathrm{x}(t)\mathrm{x}^H(t) = \frac{1}{N}\mathrm{X}\mathrm{X}^H$$

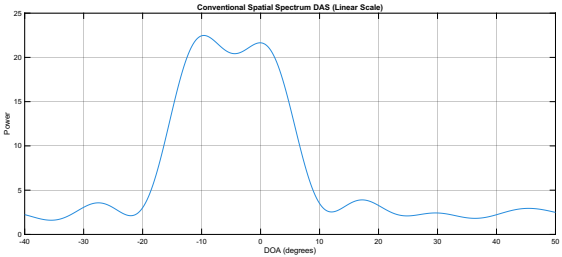
Task 1 - Sample correlation matrix



$|\hat{R}|$ shows the structure expected from a ULA.

Task 2 - DAS spatial spectrum





dB scale: Single broad peak near $\theta \approx -5^\circ$. The two sources at 0° and -10° are *not resolved*.

Linear scale: Confirms a single merged lobe.

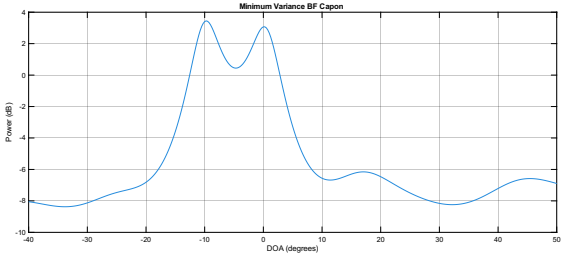
Why unresolved? The DAS beamwidth for $M = 10$, $d = \lambda/2$ is $\approx 11.5^\circ$, which exceeds the 10°.

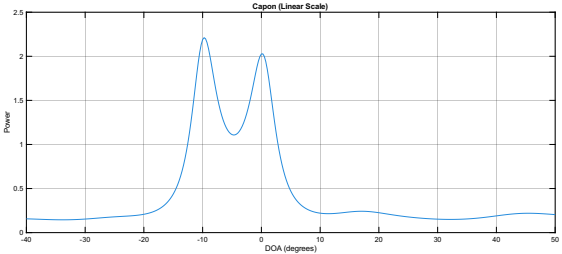
Code - DAS beamformer

```
R = x*x' / N;                    % Sample correlation matrix
d = 0.5; kd = 2*pi*d;
theta = -40:0.25:50;

P_DAS = zeros(size(theta));
for n = 1:length(theta)
    DOA = theta(n);
    phi = -kd * sin(DOA * pi / 180);
    a   = exp(1i * phi) .^ (0:M-1)';
    P_DAS(n) = (a' * R * a) / M;
end
```

$P_{\mathbf{DAS}}(\theta) = \frac{1}{M}\mathbf{a}^H(\theta)\hat{\mathbf{R}}\,\mathbf{a}(\theta)$. The conventional BF applies uniform weights. Resolution is limited by aperture size.

Task 3 - Capon spatial spectrum





dB scale: Two distinct peaks visible near 0° and -10°. The Capon BF resolves the sources that DAS could not.

Linear scale: Clear separation between the two source peaks.

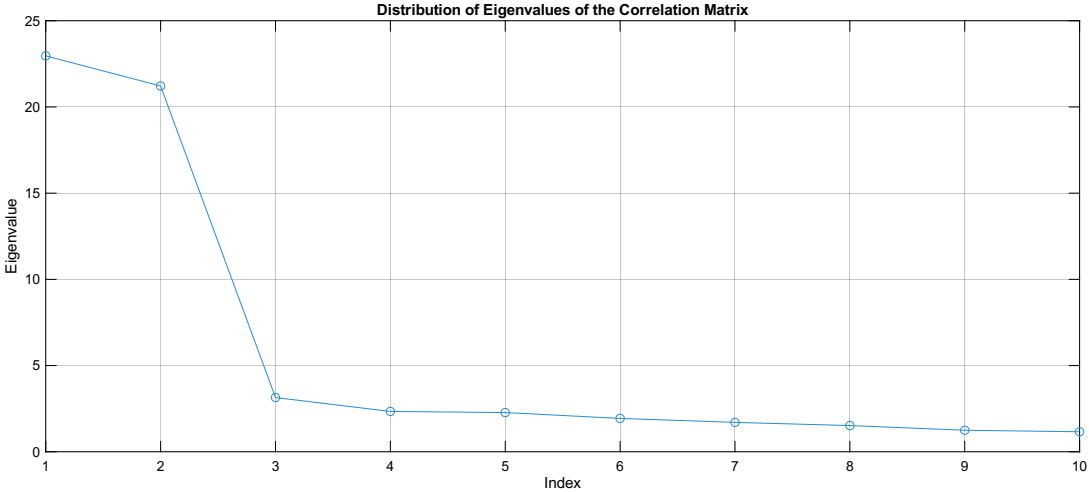
Capon (MVDR): $P_{\mathbf{Capon}}(\theta) = \frac{1}{\mathbf{a}^H(\theta)\hat{\mathbf{R}}^{-1}\mathbf{a}(\theta)}$. Minimises output power while maintaining unit gain in the look direction $\Rightarrow$ suppresses interference $\Rightarrow$ narrower peaks.

Code - Capon beamformer

```
Rinv = inv(R);
P_capon = zeros(size(theta));
for n = 1:length(theta)
    DOA = theta(n);
    phi = -kd * sin(DOA * pi / 180);
    a   = exp(1i * phi) .^ (0:M-1)';
    P_capon(n) = 1 / (a' * Rinv * a);
end
```

Data-adaptive: uses $\hat{R}^{-1}$ to place zeros on interferers. Resolution exceeds DAS but is still finite.

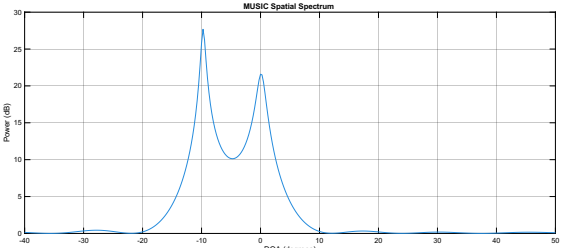
Eigenvalue distribution of $\hat{R}$

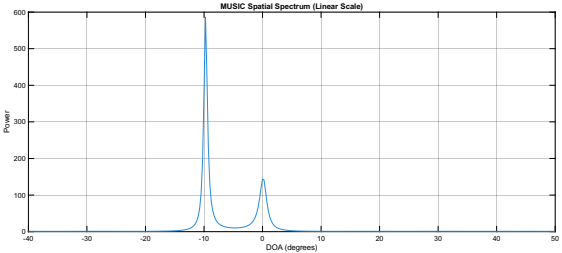


Two dominant eigenvalues $\gg$ remaining eight.

The gap between signal and noise eigenvalues enables subspace methods.

MUSIC spatial spectrum





dB scale: Sharp peaks at $\theta_1 = 0^\circ$ and $\theta_2 = -10^\circ$. MUSIC achieves super-resolution.

Linear scale: Both sources clearly resolved with very narrow peaks.

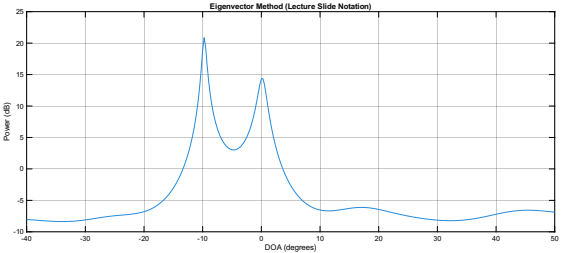
$P_{\mathbf{MUSIC}}(\theta) = \frac{\mathbf{a}^H\mathbf{a}}{\mathbf{a}^H\mathbf{U}_n\mathbf{U}_n^H\mathbf{a}}$. MUSIC exploits orthogonality between signal steering vectors and noise subspace: $\mathbf{U}_n^H\mathbf{a}(\theta_k) = 0$ at true DOAs $\Rightarrow$ denominator $\to 0 \Rightarrow$ sharp peaks.

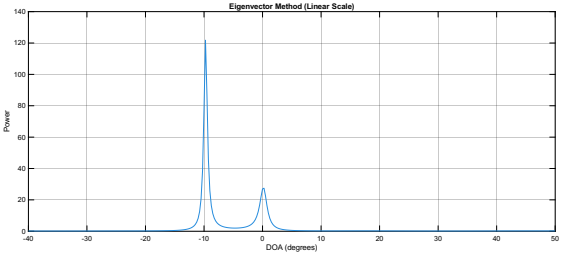
Code - MUSIC

```
[V,D] = eig(R);
[dd,I] = sort(diag(D));
dd = flipud(dd); V = V(:,flipud(I)); % descending order

num_signals = 2;
Un = V(:, num_signals+1 : end); % noise subspace
Pn = Un * Un';

P_music = zeros(size(theta));
for n = 1:length(theta)
    DOA = theta(n);
    phi = -kd * sin(DOA * pi / 180);
    a  = exp(1i * phi) .^ (0:M-1)';
    P_music(n) = (a' * a) / (a' * Pn * a);
end
```

Signal subspace: eigenvectors for the $K=2$ largest eigenvalues. Noise subspace: remaining $M-K=8$ eigenvectors. The spectrum peaks where $\mathrm{a}(\theta)$ is orthogonal to U_n.

Task 6 - Eigenvector method





dB scale: Peaks at 0° and -10°, similar to MUSIC.

Linear scale: Peak structure comparable to MUSIC with slight amplitude differences.

Eigenvector method: $P_{\mathbf{EV}}(\theta) = \frac{1}{\mathbf{a}^H(\theta)\hat{\mathbf{R}}_n^{-1}\mathbf{a}(\theta)}$, where $\hat{\mathrm{R}}_n^{-1} = \sum_{i=K+1}^{M} \frac{1}{\lambda_i}\mathrm{v}_i\mathrm{v}_i^H$.

Compared to MUSIC, the EV method weights each noise eigenvector by $1/\lambda_i$, giving more weight to eigenvectors with smaller eigenvalues, which can sharpen peaks.

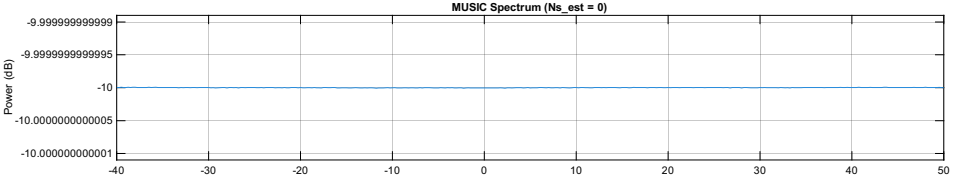
Code - Eigenvector method

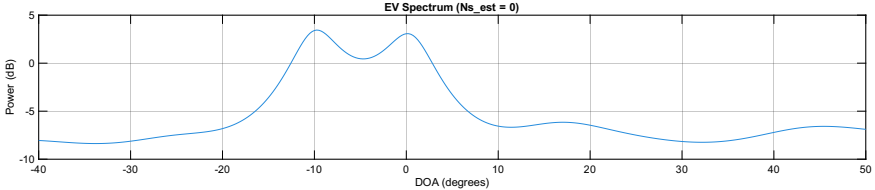
```
Vn = V(:, num_signals+1:end);
Ln = dd(num_signals+1:end);
R_noise_only_inv = Vn * diag(1 ./ Ln) * Vn';

P_EV = zeros(size(theta));
for n = 1:length(theta)
    DOA = theta(n);
    phi = -kd * sin(DOA * pi / 180);
    a   = exp(1i*phi).^(0:M-1)';
    P_EV(n) = 1 / (a' * R_noise_only_inv * a);
end
```

Key difference from MUSIC: each noise eigenvector weighted by $1/\lambda_i$ instead of unity. Both methods rely on correct estimation of K number of sources.

Task 7 - Effect of incorrect $\hat{K}$ on MUSIC & EV

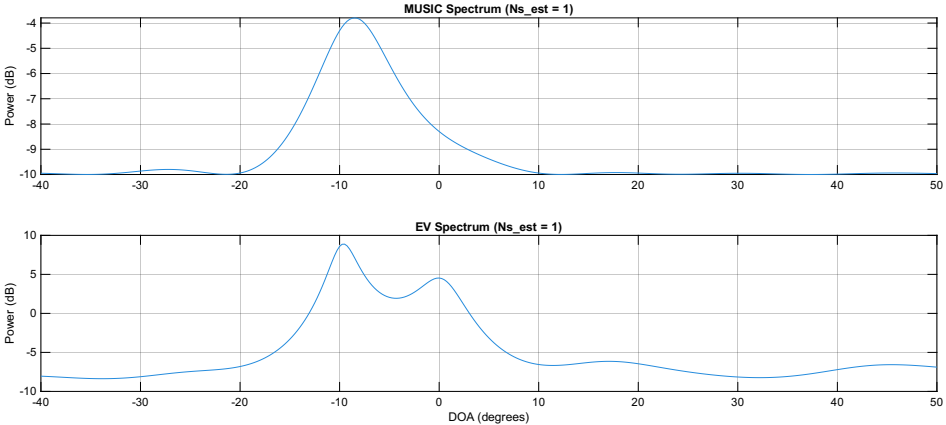
$\hat{K} = 0$ (no signal subspace. Noise subspace is the full space):





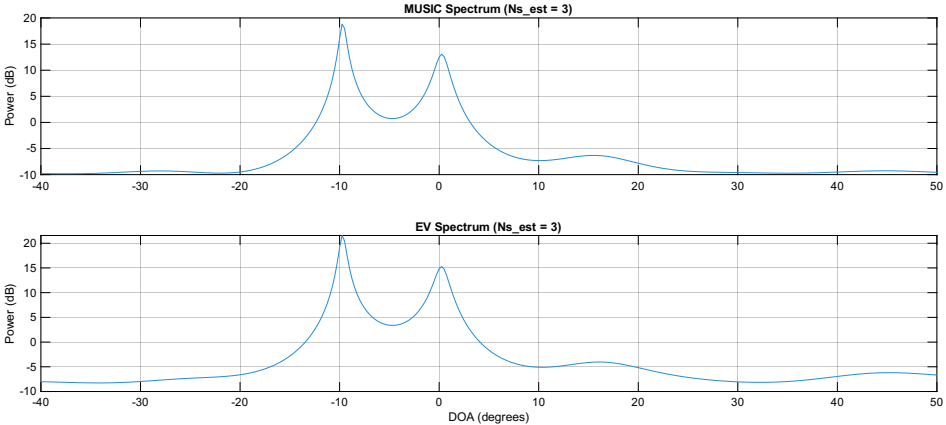
With $\hat{K}=0$ the "noise subspace" spans all M dimensions, so $\mathrm{U}_n\mathrm{U}_n^H = \mathrm{I}$. MUSIC becomes $P = M/\|\mathrm{a}\|^2 = 1$ flat, no peaks. EV spectrum is $1/(\mathrm{a}^H\mathrm{R}^{-1}\mathrm{a}) = \mathrm{Capon}$.

Task 7 - $\hat{K} = 1$



With $\hat{K} = 1$: one signal eigenvector is misclassified as noise. Only one peak is visible; the algorithm cannot resolve both sources. The noise subspace is "contaminated" by signal energy $\Rightarrow$ the orthogonality condition $\mathbf{U}_n^H \mathbf{a}(\theta_k) = 0$ no longer holds for both DOAs.

Task 7 - $\hat{K} = 3$



With $\hat{K} = 3$: the noise subspace loses one dimension. Both true peaks remain, but a spurious third peak may appear, and the noise floor rises because the noise subspace is under represented.

Code - Incorrect source count

```
Ns_estimates = [0, 1, 3];

for k = 1:length(Ns_estimates)
    Ns_est = Ns_estimates(k);
    Vn = V(:, Ns_est+1 : end);
    Ln = dd(Ns_est+1 : end);

    for n = 1:length(theta)
        DOA = theta(n);
        phi = -kd * sin(DOA * pi / 180);
        a  = exp(1i * phi) .^ (0:M-1)';

        % MUSIC
        P_music_task7(n) = 1 / (a' * (Vn * Vn') * a);

        % Eigenvector
        P_EV_task7(n) = 1 / (a' * (Vn * diag(1./Ln) * Vn') * a);
    end
end
```

Correct $\hat{K}$ is critical: underestimation loses sources, overestimation raises the noise floor and can introduce false peaks.

Summary

1. **Correlation matrix (1)**: Structure expected from a ULA; diagonal $\approx \sigma_s^2 + \sigma_n^2$.
2. **DAS (2)**: Cannot resolve sources.
3. **Capon (3)**: Data-adaptive; resolves the two sources by suppressing interference.
4. **Eigenvalues (4)**: Two dominant eigenvalues confirm $K = 2$ sources; gap separates signal and noise subspaces.
5. **MUSIC (5)**: Super-resolution via noise-subspace orthogonality; sharp peaks at true DOAs.
6. **Eigenvector (6)**: Eigenvalue-weighted noise subspace; similar resolution to MUSIC.
7. **Incorrect $\hat{K}$ (7)**: Underestimation merges/loses peaks; overestimation raises noise floor.